

RaceDay API Endpoint Plan

1. Introduction

The RaceDay API is designed to support the RaceDay event management system for South African running, walking and cycling events.

The API will allow users to register and log in, manage their profiles, browse events, manage event categories, enrol in events and manage race results.

The system has two main roles:

- **Organiser** – manages events, categories, enrolments and results.
- **Participant** – registers for events and views their enrolments and results.

2. API Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user account in the RaceDay system.	None	FirstName, LastName, Email, Password, Role, PhoneNumber	201 Created – User registered successfully ; 400 Bad Request – Invalid information ; 409 Conflict – Email already exists
POST	/api/auth/login	Authenticates a registered user and allows access to the system.	None	Email, Password	200 OK – Login successful; 401 Unauthorized – Invalid credentials
GET	/api/users/profile	Retrieves the profile of the	Any	None	200 OK – User profile returned;

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
		currently logged-in user.			401 Unauthorized
PUT	/api/users/profile	Updates the profile information of the currently logged-in user.	Any	FirstName, LastName, PhoneNumber	200 OK – Profile updated; 400 Bad Request – Invalid information; 401 Unauthorized
GET	/api/events	Retrieves a list of upcoming RaceDay events.	None	None	200 OK – List of events returned
GET	/api/events/{id}	Retrieves information about a specific event.	None	None	200 OK – Event returned; 404 Not Found – Event does not exist
POST	/api/events	Creates a new RaceDay event.	Organiser	EventName, EventDate, Location, Description, Status	201 Created – Event created; 400 Bad Request – Invalid information; 401 Unauthorized
PUT	/api/events/{id}	Updates an existing event.	Organiser	EventName, EventDate, Location, Description, Status	200 OK – Event updated; 404 Not Found – Event does not exist;

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
DELETE	/api/events/{id}	Deletes an existing event.	Organiser	None	403 Forbidden 204 No Content – Event deleted; 404 Not Found – Event does not exist; 403 Forbidden
GET	/api/events/{eventId}/categories	Retrieves all categories belonging to a specific event.	None	None	200 OK – Categories returned; 404 Not Found – Event does not exist
POST	/api/events/{eventId}/categories	Creates a new category for an event.	Organiser	CategoryName, DistanceKM, EntryFee	201 Created – Category created; 400 Bad Request – Invalid information; 404 Not Found – Event does not exist
PUT	/api/categories/{id}	Updates an existing event category.	Organiser	CategoryName, DistanceKM, EntryFee	200 OK – Category updated; 404 Not Found – Category does not exist
DELETE	/api/categories/{id}	Deletes an existing event category.	Organiser	None	204 No Content – Category deleted; 404 Not Found

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments	Allows a participant to enrol in an event category.	Participant	CategoryID	– Category does not exist 201 Created – Enrolment created; 400 Bad Request – Invalid information ; 409 Conflict – Participant already enrolled
GET	/api/enrolments/my	Retrieves all event enrolments belonging to the logged-in participant.	Participant	None	200 OK – Enrolments returned; 401 Unauthorized 204 No Content – Enrolment cancelled;
DELETE	/api/enrolments/{id}	Cancels an existing participant enrolment.	Participant	None	404 Not Found – Enrolment does not exist
GET	/api/events/{eventId}/enrolments	Retrieves participants enrolled in a specific event.	Organiser	None	200 OK – Enrolments returned; 403 Forbidden – Access denied; 404 Not Found
POST	/api/results	Records a participant's	Organiser	EnrolmentID, FinishTime,	201 Created – Result

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
		result after an event.		Position, ResultStatus	recorded; 400 Bad Request – Invalid information ; 404 Not Found – Enrolment does not exist 200 OK – Result returned;
GET	/api/results/{id}	Retrieves a specific race result.	Any	None	404 Not Found – Result does not exist 200 OK – Results returned;
GET	/api/participants/results	Retrieves the logged-in participant's performance history.	Participant	None	401 Unauthorized 200 OK – Result updated;
PUT	/api/results/{id}	Updates an existing participant race result.	Organiser	FinishTime, Position, ResultStatus	400 Bad Request – Invalid information ; 404 Not Found

3. Role Summary

Organiser

An Organiser can:

- Create events.
- Update events.
- Delete events.

- Create event categories.
- Update categories.
- Delete categories.
- View event enrolments.
- Record participant results.
- Update participant results.

Participant

A Participant can:

- Register for an account.
- Log into the system.
- View and update their profile.
- Browse upcoming events.
- View event categories.
- Enrol in event categories.
- View their enrolments.
- View their race results and performance history.
- Cancel their enrolments.

Public Access

Users who are not logged in can:

- Register for an account.
- Log into the system.
- View upcoming events.
- View individual event details.
- View event categories.

4. HTTP Methods

The API will use standard RESTful HTTP methods:

- **GET** – Retrieve information.
- **POST** – Create new information.
- **PUT** – Update existing information.
- **DELETE** – Remove information.

5. Conclusion

The endpoint plan provides the structure that will be followed when the RaceDay RESTful API is developed in Part 2. The planned endpoints cover authentication, user profiles, events, categories, event enrolments and results as required by the project specification.

